Chapter 4 The Estimation of Sample Size

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A hypothetical example

An anthropologist wish to estimate the percentage of inhabitants belonging to blood group O.

- A simple random sample
- An important question: How large should the sample be?
- Two preliminary questions needed
  - How accurately does the anthropologist wish to know the percentage of people with blood group O?
  - How large is the chance that the anthropologist is willing to take of getting a very unlucky sample?

In technical terms:

The sample percentage p is to lie in the range $(P \pm 5)$, except for a 1 in 20 chance.

- Suppose p is normally distributed, and the fpc is ignored
- p : lie in the range $(P \pm 1.96\sigma_p)$, apart from a 1 in 20 chance

$$\sigma_p \doteq \sqrt{PQ/n}$$

$$1.96\sqrt{PQ/n} = 5 \qquad \text{or} \qquad n = \frac{1.96^2 PQ}{25}$$

- What is the value of P?
- The anthropologist: P is likely to lies in the range 30 to 60%.
- The size n lies between 323 and 384 (P=50%).

Analysis of the problem

- Some statement concerning what is expected of the sample.
- Some equation that connects n with the desired precision of the sample
- Unknown parameters in the equation to be estimated

- For each subdivision: data and desired limits of error
- More than one item or characteristic in a survey
- The chosen value of n $\Leftrightarrow$ the resources available

The formula for n in sampling for proportions

- We want $P(|p - P| \geq d) = \alpha$
- Assume: SRS, normal distribution
- The standard error of p is $\sigma_p = \sqrt{\frac{N-n}{N-1}}\sqrt{\frac{PQ}{n}}$
- The formula that connects n with the desired degree of precision is

$$d = t\sqrt{\frac{N-n}{N-1}}\sqrt{\frac{PQ}{n}}$$

- Solving for n:

$$n = \frac{\frac{t^2PQ}{d^2}}{1 + \frac{1}{N}\left(\frac{t^2PQ}{d^2} - 1\right)}$$

- Substitute an advance estimate p of P:

$$n = \frac{\frac{t^2 pq}{d^2}}{1 + \frac{1}{N}\left(\frac{t^2 pq}{d^2} - 1\right)}$$

- Another expression

$$n_0 = \frac{t^2 pq}{d^2} = \frac{pq}{V} \qquad n = \frac{n_0}{1 + (n_0 - 1)/N}$$

where $V = (d/t)^2$ is the desired variance of p.



Example

In the hypothetical blood groups example,

$$d = 0.05, \quad p = 0.5, \quad \alpha = 0.05, \quad t = 1.96$$

We have

$$n_0 = \frac{t^2 pq}{d^2} = \frac{1.96^2 \times 0.5 \times 0.5}{0.0025} = 384$$

Assume that there are only 3200 people on the island. The fpc is needed,

$$n = \frac{n_0}{1 + (n_0 - 1)/N} = \frac{384}{1 + 383/3200} = 342$$

- Wish to control the relative error r instead of the absolute error d
- For example, wish to estimate NP with an error not exceeding 10%

$$P\left(\frac{|\,Np-NP\,|}{NP}\geq r\right)=P(|\,p-P\,|\geq rP)=\alpha$$

- Need to do: substitute rP or rp for d in formulas

$$n_0=\frac{t^2pq}{d^2}=\frac{pq}{V} \quad \Rightarrow \quad n_0=\frac{t^2pq}{r^2p^2}=\frac{t^2q}{r^2p}$$

$$n=\frac{n_0}{1+(n_0-1)/N}$$

The formula for n with continuous data

- Wish to control the relative error r
- With a simple random sample having mean $\bar{y}$,

$$P\left(\frac{|\bar{y}-\bar{Y}|}{\bar{Y}} \geq r\right) = P\left(\frac{|N\bar{y}-N\bar{Y}|}{N\bar{Y}} \geq r\right) = P(|\bar{y}-\bar{Y}| \geq r\bar{Y}) = \alpha$$

- Assume $\bar{y}$ is normally distributed, the standard error is

$$\sigma_{\bar{y}} = \sqrt{\frac{N-n}{N}}\frac{S}{\sqrt{n}} \qquad r\bar{Y} = t\sigma_{\bar{y}} = t\sqrt{\frac{N-n}{N}}\frac{S}{\sqrt{n}}$$

$$\Rightarrow \quad n = \left(\frac{tS}{r\bar{Y}}\right)^2 \Bigg/ \left[1+\frac{1}{N}\left(\frac{tS}{r\bar{Y}}\right)^2\right]$$

The coefficient of variation

$$n = \left(\frac{tS}{r\bar{Y}}\right)^2 \Bigg/ \left[1 + \frac{1}{N}\left(\frac{tS}{r\bar{Y}}\right)^2\right]$$

- The size n depends on the coefficient of variation $S / \bar{Y}$.
- Need to guess in advance, more stable and easier
- Another expression

$$n_0 = \left(\frac{tS}{r\bar{Y}}\right)^2 = \frac{1}{C}\left(\frac{S}{\bar{Y}}\right)^2 \qquad n = \frac{n_0}{1 + n_0 / N}$$

The quantity $C = \left(\frac{r}{t}\right)^2 = \left(\frac{r\bar{Y}}{t}\frac{1}{\bar{Y}}\right)^2$ is the desired $(\mathrm{cv})^2$ of $\bar{y}$.

- Given the absolute error d in $\bar{y}$

$$n_0 = \frac{t^2 S^2}{d^2} = \frac{S^2}{V} \qquad n = \frac{n_0}{1 + n_0 / N}$$

where V is the desired variance of $\bar{y}$.



Example

The data that follow were obtained from a bed of silver maple seedlings 1 ft wide and 430 ft long. The sampling unit was 1 ft of the length of the bed, so that $N = 430$. By complete enumeration of the bed if was found that $\bar{Y} = 19, S^2 = 85.6$, these being the true population values.

With simple random sampling, how many units must be taken to estimate $\bar{Y}$ within 10%, apart from a chance of 1 in 20?

$$n_0 = \frac{t^2 S^2}{r^2 \bar{Y}^2} = \frac{1.96^2 \times 85.6}{0.1^2 \times 19^2} = 91$$

Since n_0 / N is not negligible, $n = \dfrac{91}{1 + 91/430} = 75$

Advance estimates of population variances

- By taking the sample in two steps, the first being a simple random sample of size n_1 from which estimates s^2 or p_1 of S^2 or P and the required n will be obtained;
- By the results of a pilot survey;
- By previous sampling of the same or a similar population;
- By guesswork about the structure of the population, assisted by some mathematical results.

Sample size with more than one item

- **One method**

  1. Identify the most vital items for the survey
  2. Specify margins of error for the items
  3. Estimate the sample size needed for each of these important items

- If the n's are all reasonably close, and the largest of the n's falls within the limits of the budget, select the largest.
- If there is a sufficient variation among the n's, we may use a smaller value of n. (budgetary consideration or an too high over-all standard of precision)

Sample size in decision problems

- Determine sample size when a practical decision is to be made from the results of the sample.
- $l(z)$: the loss incurred in a decision through an error z
- $f(z,n)$: the frequency distribution of z
- Expected loss:

$$L(n) = \int l(z) f(z,n) dz$$

- A reasonable procedure is to choose n to minimize

$$C(n) + L(n)$$

where $C(n)$ is the cost of a sample of size n.

- Alternatively, use the term gain that accrues from having the sample information
- Expected gain: $G(n)$
- We choose n to maximize $G(n) - C(n)$



Example The loss function is $l(z) = \lambda z^2$. The expected loss is $L(n) = \lambda E(z^2)$. If $\hat{\bar{Y}}$ is the sample estimate of $\bar{Y}$, and $z = \hat{\bar{Y}} - \bar{Y}$

$$L(n) = \lambda V(\hat{\bar{Y}}) = \frac{\lambda S^2}{n} - \frac{\lambda S^2}{N}$$

Consider the simplest cost function: $C(n) = c_0 + c_1 n$

By differentiation: $n = \sqrt{\lambda S^2 / c_1}$

The design effect (deff)

- The design effect (deff) of the plan

$$deff = V_d(\hat{\theta}) / V_{srs}(\hat{\theta})$$

$$deff = \mathrm{MSE}_d(\hat{\theta}) / \mathrm{MSE}_{srs}(\hat{\theta})$$

- Notice: the same number of units in the two samples!
- Two uses: (1) in appraising the efficiency of more complex plans; (2) in sample-size estimation.
- A simple random sample of person: $v_{srs}(p) = 0.00197$

  Another complex sampling design: $v_d(\hat{P}) = 0.00520$

$$deff = \frac{0.00520}{0.00197} = 2.6$$

- When the sampling fractions are small, we can estimate sample size of the complex plan by calculating the n (number of persons) needed with a simple random sample of persons and multiplying by 2.6.

- Judge whether the complex plan is advantageous in efficiency relative to its cost and complexity, in terms of either the value of deff and the sample size.